



$$-(r_j - r_i)^\top M(r_j - r_i)$$

geo.pos3d

$$\mathbb{1}[j \text{ ancestor of } i]$$

kin.ancestor

$$\text{flow}^+(\text{support})(i, j)$$

ix.force_flow

$$g_{\text{task next}(i, j)}$$

task.next_contact

$$\text{logit}(i, j) = \frac{\langle q_i, k_j \rangle}{\sqrt{d}} + b_{\text{dist}} + b_{\text{dir}} + b_{\text{kin}} + b_{\text{contact}} + b_{\text{supp/flow}} + b_{\text{held}} + g_{\text{task}} b_{\text{next}} + b_{\text{ui}}$$

learned, data-dependent (no heat)

geo.pos3d

$$\tilde{b}^\top(r_j - r_i)$$

ix.contact

$$\text{contact}(i, j)$$

ix.held_by

$$\text{held_by}(i, j)$$

ui.label_for

$$\text{label_for}(i, j), \text{above}(i, j)$$

